READING 2.3 · SHIP AND OCEAN SYSTEMS DESIGN

The Admiralty Law

Power, speed and displacement for one hull form: the physics, the deduction, its limits, parent ships and a worked example

Papanikolaou (2026) §1.3, §2.3, §2.14 · Session W2.3 · extended reading after Lecture 2.3a, not for evaluation

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Where the Admiralty Law Comes From

1

$$R_T = \tfrac{1}{2}\,\rho\,S\,V^2\,C_T \qquad P_E = R_T\,V, \qquad P_B = \frac{P_E}{\eta_D\,\eta_S}$$

Resistance; then effective and brake power through the propulsive and shaft efficiencies

2

$$S = C_S\sqrt{\nabla L} = C_S\sqrt{L/\nabla^{1/3}}\;\nabla^{2/3} = k_S\,\nabla^{2/3}, \qquad \Delta = \rho\,\nabla$$

Taylor (Eq. 2.8)[1]: one hull shape fixes CS and L/∇ to the 1/3, so kS is constant

3

$$P_B = \frac{R_T\,V}{\eta_D\,\eta_S} = \frac{\tfrac{1}{2}\,\rho\,k_S\,\nabla^{2/3}\,C_T\,V^3}{\eta_D\,\eta_S} = \frac{\rho^{1/3}\,k_S\,C_T}{2\,\eta_D\,\eta_S}\;\Delta^{2/3}\,V^3$$

Substitute 2 into 1, with ∇ = Δ/ρ: brake power grows with Δ to the 2/3 and with V³

4

$$C = \frac{\Delta^{2/3}\,V^3}{P_B} = \frac{2\,\eta_D\,\eta_S}{\rho^{1/3}\,k_S\,C_T}$$

Hold CT, ηD, ηS, kS (same hull, Fn, Rn): C is constant (2.104), on PB

Symbols

R_T	total resistance [kN]
ρ	sea water density, 1.025 t/m³
S	wetted surface [m²]
V	speed [m/s]; [kn] in C
C_T	total resistance coefficient
P_E, P_B	effective, brake power [kW]
η_D, η_S	propulsive, shaft efficiency
∇	displaced volume [m³]
Δ	displacement [t]
C_S	Taylor coefficient, f(B/T, CM)
k_S	shape constant of the hull
C	Admiralty constant, typically 400–600 with Δ [t], V [kn], P [kW]

[1] Taylor’s formula estimates the wetted surface from ∇ and L alone, before any lines exist, for the frictional resistance; CS is read against B/T and CM (Lewis 1988). It suits conventional displacement hulls; for a bulb, transom or unusual sections use the lines, or the S of Holtrop–Mennen (Tool 2.4).

Power, Speed and Displacement: Three Relations

Power for a new speed or displacement

$$\frac{P_2}{P_1} = \left(\frac{\Delta_2}{\Delta_1}\right)^{2/3} \left(\frac{V_2}{V_1}\right)^{3}$$

Speed for a new power or displacement

$$\frac{V_2}{V_1} = \left(\frac{P_2}{P_1}\right)^{1/3} \left(\frac{\Delta_1}{\Delta_2}\right)^{2/9}$$

Displacement for a new power or speed

$$\frac{\Delta_2}{\Delta_1} = \left(\frac{P_2}{P_1}\right)^{3/2} \left(\frac{V_1}{V_2}\right)^{9/2}$$

Same ship, same C: write 2.104 for state 1 and state 2, and divide.

Change one by +10 %, hold another: the third follows

Hold	Change	Result
Δ	V +10 %	P +33.1 %
V	Δ +10 %	P +6.6 %
P	Δ +10 %	V −2.1 %
Δ	P +10 %	V +3.2 %
P	V +10 %	Δ −34.9 %

Speed is expensive, displacement is cheap: power goes with V³ but only with Δ to the 2/3.

The book's use (§1.3): C measured on sea trials at a reduced draught gives the speed at the design displacement with the installed power.

$$P_2 = P_1 \left(\frac{\nabla_2}{\nabla_1}\right)^{2/3}$$

The sea-trial standard uses the same law: ITTC 7.5-04-01-01.1, Speed/Power Trials (revision 08, 2024)

· §10.3.5 corrects the measured trial power for a different displacement, "based on the Admiralty formula", at the same speed

· §5.1 allows this only if the trial displacement is within 2 % of the required one

Order of Resistance, Order of Power, and the Humps

$$R \propto V^{\,n} \;\Rightarrow\; P = RV = a\,V^{\,m}, \qquad m = n + 1$$

$$C = \frac{\Delta^{2/3}\,V^{3}}{P} = \frac{\Delta^{2/3}\,V^{3}}{a\,V^{\,m}} = \frac{\Delta^{2/3}}{a}\;V^{\,3-m}$$

a: a constant of the ship. Δ fixed: C constant only if m = 3

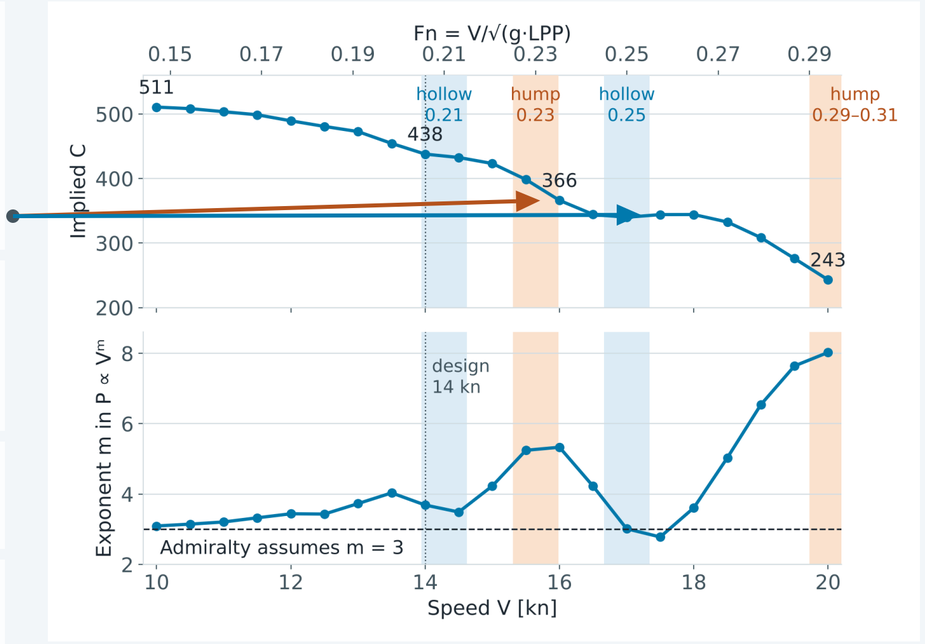
m > 3: C falls (hump) · **m < 3: C rises** (hollow)

$$C_F = \frac{0.072}{R_n^{0.2}} \;\Rightarrow\; R_F \propto V^{1.8}, \quad P_F \propto V^{2.8}$$

Friction alone (Prandtl, Eq. 2.9): m just below 3.

Waves: CW swings with Fn (Fig. 2.14). Towards a hump m rises above 3 and C falls; in a hollow m drops back to about 3.

Reference ship: m 3.1 at 10 kn; above 5 at 15.5–16 kn, on the 0.23 hump; 2.8 at 17.5 kn, in the 0.25 hollow. C falls from 511 to 243.



Range of validity, read from the chart: one C serves only where m ≈ 3

Same C across speeds? Within about ±1 kn

10–12 kn: 511 to 489 (−4 %) · 13.5–15 kn: 454 to 423 (±4 % of 438).

Not across a hump

15–16.5 kn, Fn 0.22–0.24: C drops 19 % (423 to 344); above Fn 0.27, again.

A hollow has its own C

16.5–18 kn, Fn 0.25: C flat at 340–344, but not the 438 of 14 kn.

Getting C from Parent Ships

1. Collect parents in D2: same type, similar L, Fn and CB.
2. One power definition for all: MCR, SMCR, service PB or trial PD.
3. C for each; use the closest parent, not the 400–600 range.
4. The spread of C is your uncertainty: carry it into the power and the machinery weight.

Source	Δ [t]	V [kn]	P [kW]	Basis	C
Lecture 1.5, assumed	13,200.7	14	3,832	installed	400
Holtrop–Mennen	13,200.7	12	1,972	SMCR	489
Holtrop–Mennen	13,200.7	14	3,500	SMCR	438
Holtrop–Mennen	13,200.7	16	6,248	SMCR	366
Parent 1 · your D2	…	…	…	…	…
Parent 2 · your D2	…	…	…	…	…

The reference ship rows: Lecture 1.5 and Tool 2.4. One hull already spans C 366–489 between 12 and 16 kn.

Units: P in metric hp gives C(kW)/1.36; V in m/s gives another constant.

Basis: SMCR = 1.28·PB (15 % margin, 90 % MCR), so C on PB is 1.28 × C on SMCR.

Variants: Völker (2.105) uses V^3.6; Heickel (2.106) a similar coefficient.

One Constant, a Domain of Ships

$$C = \frac{\Delta_0^{2/3}\, V_0^3}{P_0} = \frac{13\,200.7^{2/3} \cdot 14^3}{3\,500} = 438$$

Parent: the reference ship at 14 kn, SMCR by Holtrop–Mennen. Δ changed through the draught, L and B fixed. Each cell: Admiralty / Holtrop–Mennen.

SMCR [kW] for Δ and V

Δ / Δ0	13 kn	14 kn	15 kn
0.90	2,612 / 2,433	3,263 / 3,272	4,013 / 4,152
1.00	2,803 / 2,596	3,500 / 3,500	4,305 / 4,454
1.10	2,986 / 2,753	3,730 / 3,723	4,588 / 4,750

Speed [kn] on the same 3,500 kW

Δ / Δ0	Admiralty / H–M
0.80	14.71 / 14.62
0.90	14.33 / 14.28
1.00	14.00 / 14.00
1.05	13.85 / 13.88
1.10	13.71 / 13.78

Δ scales well: within 1 % at 14 kn, ±10 % of Δ.

V scales only near the parent: +8 % at 13 kn, −3 % at 15 kn, because m ≠ 3.

Design loop: 5 % more Δ costs 3.3 % power at 14 kn, or 0.15 kn on the same engine.